Chenxu Liu 刘宸旭

B.Sc. in Physics — Shandong University, Qingdao

202422201069@mail.sdu.edu.cn · physicsliu299792458@outlook.com · github.com/PhyLiusir
Qingdao, China

EDUCATION

Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Coursework: theoretical mechanics, electrodynamics, thermodynamics and statistical physics, quantum mechanics, advanced quantum mechanics, quantum field theory, particle physics, general relativity.

RESEARCH INTERESTS

I am interested in how much of hadron structure can be derived rather than fitted.

Light-front QCD. Quantise on the light front and the vacuum turns trivial — the difficulty does not disappear, it reappears in the Fock truncation and in the zero modes the $A^+=0$ gauge leaves behind. Next for me: the holographic link from confinement scale to bound-state equation.

Hadron structure from Fock wave functions. A transverse-momentum distribution read as a spectator-integrated amplitude, not a fitting function, with $\Sigma x_i=1$ and $\Sigma \mathbf{k}_{\perp i}=0$ built in from the start.

TMD factorisation. Collins–Soper evolution and its non-perturbative kernel; the SIDIS structure-function dictionary (Sivers, Collins, Boer–Mulders); the Drell–Yan sign reversal that keeps SIDIS and DY from being fitted apart.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Light-front quantisation (PH 01).** Working toward: an answer to where the Fock-space truncation can still be trusted. The two handles I am following are DLCQ matrices whose UV cutoff is matched to the truncation, and what the zero modes keep carrying once the gauge is fixed.
- **Baryon light-cone Fock expansion (PH 02).** Working toward: a truncation checkable against constraints that do not bend — normalisation ($\Sigma_n \int |\psi_n|^2=1$, $\int dx\, f_1$ equal to the valence count) and the mass relation $P^+P^- - P_\perp{}^2 = M^2$ — three-quark sector first, then five, and which observables survive a cut at fixed N.
- **Proton structure — DIS / SIDIS / DY (PH 03).** Working toward: one picture the three processes agree on. TMD factorisation at $P_{hT} \ll Q$, matched onto a Y-term at large transverse momentum, Collins–Soper evolution with a non-perturbative kernel, and where a Gaussian width in $\langle k_\perp{}^2 \rangle$ gives way to power-law tails.

OPEN QUESTIONS

- Does convergence come from restricting $k_\perp$ or from the number of Fock components?
- Do SIDIS and DY disagree on the same $\langle k_\perp{}^2 \rangle$ — and is a disagreement a sign reversal or an evolution effect?
- In which Q^2 does a Gaussian width stop being defensible, i.e. where do non-Gaussian tails take over?
- Can a kernel fitted to Drell–Yan alone predict the SIDIS P_{hT} spectrum with no new parameters?

TECHNICAL SKILLS

LANGUAGES	Python, C++, Bash, Git
SYMBOLIC	Mathematica for algebra; SymPy — mathkit, an open-source skill I maintain; numerical integration and fitting
THEORY	Light-front quantisation, Fock-basis expansion, TMD factorisation and evolution, PDF and TMD fits, perturbative calculations
TOOLING	LaTeX and TikZ, reproducible notes, Linux and HPC

LANGUAGES

Chinese (native) · English (working proficiency).